Measures of Centre and Spread

Data Analysis

2026

Preliminary Mathematics Standard

Outline

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What is a Measure of Centre?

A **measure of centre** describes a typical or central value in a dataset.

There are three main measures:

- **Mean** — the arithmetic average
- **Median** — the middle value when data is ordered
- **Mode** — the most frequently occurring value

Mean

Definition: The sum of all values divided by the number of values.

$$\bar{x} = \frac{\sum x}{n}$$

- Uses every value in the dataset
- Affected by outliers

Example

Dataset: $4,\ 7,\ 3,\ 9,\ 2,\ 7,\ 6$

$$\bar{x} = \frac{4+7+3+9+2+7+6}{7} = \frac{38}{7} \approx 5.43$$

Median

Definition: The middle value of an ordered dataset.

Finding the median:

- Order data from smallest to largest
- If n is **odd**: median is the $\dfrac{n+1}{2}$th value
- If n is **even**: median is the average of the $\dfrac{n}{2}$th and $\left(\dfrac{n}{2}+1\right)$th values

Example
Ordered dataset: $2,\ 3,\ 4,\ \mathbf{6},\ 7,\ 7,\ 9 \quad (n = 7, \text{odd})$

$$\text{Median} = \text{4th value} = 6$$

Mode

Definition: The value that appears most often in a dataset.

- A dataset can have **no mode**, **one mode**, or **multiple modes**
- The only measure of centre usable with **categorical data**
- Not affected by outliers

Example
Dataset: $2,\ 3,\ 4,\ 6,\ 7,\ 7,\ 9$

$$\text{Mode} = 7 \quad \text{(appears twice)}$$

Calculating from a Frequency Table

Value (x)	Freq (f)
2	3
4	5
6	2
8	4
Total	14

Example

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{2(3) + 4(5) + 6(2) + 8(4)}{14}$$
$$= \frac{70}{14} = 5$$

Mode $= 4$ (highest frequency)

Median: average of 7th and 8th values — both in the "4" group

$\therefore$ Median $= 4$

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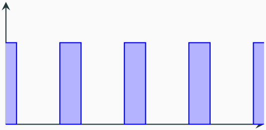
Distribution Shapes

Datasets are described by the number of peaks they have:

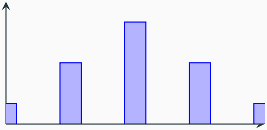
- **Uniform** — all values occur with roughly equal frequency; no clear peak
- **Unimodal** — one clear peak
- **Bimodal** — two peaks of roughly equal height
- **Multimodal** — more than two peaks

Distribution Shapes — Diagrams

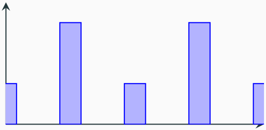
Uniform



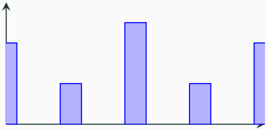
Unimodal



Bimodal



Multimodal



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What is a Measure of Spread?

A **measure of spread** describes how varied or dispersed the values in a dataset are.

Two key measures:

- **Range** — simplest measure; difference between maximum and minimum
- **Standard deviation** — how far values typically deviate from the mean

A **small** spread $\Rightarrow$ values are close together.
A **large** spread $\Rightarrow$ values are more variable.

Range

Definition:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

- Simple to calculate
- Only uses two values — heavily affected by outliers
- Does not describe how the middle data is distributed

Example
Dataset: $2,\ 3,\ 4,\ 6,\ 7,\ 7,\ 9$

$$\text{Range} = 9 - 2 = 7$$

Population Standard Deviation

Definition

The average distance of each value from the mean.

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

Example

Dataset: $3,\ 5,\ 7,\ 7,\ 8 \quad (\bar{x} = 6)$

x	$x - \bar{x}$	$(x - \bar{x})^2$
3	-3	9
5	-1	1
7	$+1$	1
7	$+1$	1
8	$+2$	4
		$\sum = 16$

$$\sigma = \sqrt{\frac{16}{5}} = \sqrt{3.2} \approx 1.79$$

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Comparing Datasets

To compare two datasets, examine both their **centre** and **spread**.

Example

	Dataset A	**Dataset B**
Values	$4, 5, 6, 6, 7, 8$	$1, 3, 6, 8, 10, 12$
Mean	6.00	6.67
Median	6.00	7.00
Range	4	11
σ	1.29	3.86

Both datasets have similar centres, but Dataset B has a much larger spread — its values are far more variable.

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Merits of Each Measure of Centre

Measure	Use when...	Avoid when...
Mean	Data is symmetric with no outliers	Outliers or skewed data present
Median	Outliers or skewed data present	A measure using all values is needed
Mode	Categorical data; finding the most common value	Data has no repeated values

Real-World Examples

Appropriate uses:

- **Median** for house prices — a few very expensive homes skew the mean
- **Mode** for shoe sizes — most common size determines stock orders
- **Mean** for exam scores in a balanced class — gives a fair overall picture

Inappropriate uses:

- **Mean** for income data — billionaires distort the average upward
- **Mode** for continuous data (e.g. heights) — values rarely repeat exactly
- **Range** alone for spread — one extreme value makes it misleading

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Practice — Measures of Centre

Example

The following data shows the number of hours 8 students studied last week:

$$3,\ 5,\ 5,\ 7,\ 8,\ 9,\ 9,\ 12$$

1. Calculate the **mean**.
2. Find the **median**.
3. State the **mode**.
4. Describe the **shape** of this distribution.

Practice — Frequency Table

Example

Score	Frequency
10	2
20	6
30	4
40	3
50	1

Using the frequency table:

1. Calculate the **mean score**.
2. Find the **median score**.
3. State the **mode**.

Example

Two athletes record their sprint times (seconds) over 6 trials:

Athlete A: $11.2,\ 11.4,\ 11.3,\ 11.5,\ 11.2,\ 11.4$

Athlete B: $10.8,\ 12.1,\ 11.0,\ 11.9,\ 10.5,\ 12.7$

1. Calculate the **mean** for each athlete.
2. Calculate the **range** for each athlete.
3. Calculate the **population standard deviation** for each athlete.
4. Which athlete is more **consistent**? Justify your answer.

Practice — Choosing the Right Measure

Example

1. A real estate agent reports the mean house price in a suburb as \$1 250 000. Most houses sell for around \$800 000, but one recently sold for \$6 500 000.
   (a) Explain why the mean is misleading here.
   (b) Which measure of centre would be more appropriate? Why?

2. A café sells 5 sizes of coffee. Last week's sales were: Small (12), Medium (45), Large (38), Extra Large (20), Jumbo (5).
   (a) Which measure of centre is most appropriate?
   (b) State its value.

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Homework

Complete all questions from *Data Analysis — Measures of Centre and Spread*.